

Item 3b: Model Comparison & Evaluation

Below is the evaluation of the 4 models when I asked the question *"Is this course hard?"* using the course syllabus document:

1. Output Summaries & Results

- **gpt-3.5-turbo:**
 - **Behavior:** Returned a very high-level, generic paragraph about Python and AI concepts. It lacked specific syllabus details (like grading distribution, lab deadlines, or project requirements).
- **gpt-4.1:**
 - **Behavior:** Returned a highly structured, accurate, and comprehensive response. It cited specific policies directly from the syllabus (e.g., Thursday lab deadlines, Sunday homework deadlines, 25% exam, 25% group project, \$10–\$30 API cost estimate) and provided a balanced verdict based on prior Python knowledge.
- **gpt-5-chat-latest:**
 - **Behavior:** Returned an API error (404 - model_not_found / deprecated). This model alias is no longer available in the OpenAI API.
- **gpt-5-nano:**
 - **Behavior:** Returned a fast, well-structured response with clear bullet points ("Why it's challenging", "Who might find it harder", "Tips to succeed"). It captured key specifics (9 labs, 7 homeworks, team project details) accurately.

Which Answer is Best? (Quality, Cost & Speed Evaluation)

- **Best Overall Quality Answer: gpt-4.1**
 - **Reasoning:** gpt-4.1 provided the most precise synthesis of the syllabus. It correctly identified every nuance—from grading weights and late policies to specific technical concepts (Streamlit, RAG, APIs)—and gave a clear, actionable conclusion for the student.
- **Best Answer Considering Cost, Speed, and Efficiency: gpt-5-nano**
 - **Reasoning:** gpt-5-nano is significantly faster and lower cost per query than gpt-4.1. Despite its smaller footprint, it extracted all critical syllabus facts

(lab counts, homework counts, project percentages, API costs) and formatted them concisely without missing key details.

- **Conclusion:**

gpt-4.1 is the best model for overall quality and depth of analysis. However, when factoring in **cost and speed**, **gpt-5-nano** is the best choice for everyday Q&A applications because it delivers nearly identical accuracy at a fraction of the response time and latency cost. (gpt-3.5-turbo lacked sufficient detail, and gpt-5-chat-latest failed due to API deprecation).

References:

1. **Gemini:** Used understand to understand code and make changes according to get the desired output.